AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter Monitoring & Health Effects

Issue 2026-08-11 | Entry window 11 August 2026 | 26 records carried, 24 rejected

26 RECORDS CARRIED (24 REJECTED)	**8** SUBTOPIC CLUSTERS OF TEN REPRESENTED	**7** RECORDS FROM CHINA (27% OF CORPUS)	**7** EFFECT ESTIMATES WITH INTERVALS	**6** METADATA-ONLY RECORDS

Signal of the day

A national PurpleAir calibration that uses only sensor-reported parameters — no external meteorology, no ancillary fields — reaches $r = 0.90$–0.96 against EPA reference monitors on a held-out year.

Balasubramanian et al. (*Environ Pollut*) train a random-forest correction for PurpleAir $PM_{2.5}$ across the United States on 2019–2021 data and evaluate it on **2022**, a genuine temporal hold-out rather than a random split. The design choice that matters is the feature set: PM_1, $PM_{2.5}$, PM_{10}, temperature, pressure and relative humidity, *all reported by the sensor itself*. Nothing is drawn from a reanalysis product or a nearby met station. Pearson r against co-located EPA monitors is **0.90–0.96** across buffer radii from 500 m to 5 km. Two things follow for distributed sensing. First, a self-contained feature set is what makes a correction deployable at the edge and in regions with no regulatory infrastructure — the stated motivation, and the reason this is more useful than a better-performing model that needs a meteorological feed. Second, the buffer sweep is an implicit spatial-transferability experiment: performance holding from 500 m to 5 km is the quantity a network planner actually needs, and it is reported here as a design parameter rather than a limitation. The caveat is that r is not an error metric. No RMSE, bias or slope is quoted in the abstract, and r is insensitive to exactly the multiplicative bias that humidity-driven hygroscopic growth produces. A correction can be strongly correlated and still wrong by a factor. Until the error statistics are read off the paper, treat the headline as evidence of *tracking*, not of accuracy.

What changed today

- **The forest plot is back after seven empty issues.** Seven interval estimates were extractable today, against zero on each of 4–10 August. Both sources are Chinese: a 24,333-participant thyroid cross-section and a six-year Xi'an outpatient time series. The thyroid result — OR **1.27** (**95% CI 1.06–1.51**) for subclinical hypothyroidism per 10 $\mu g/m^3$ $PM_{2.5}$, with FT4 down 3.57% and TSH up 12.15% — is the day's only well-powered dose–response on a novel endpoint.
- **A negative result about lockdowns that deserves attention.** Wang et al. find combustion-source pollutant–respiratory associations were *stronger* during strict COVID control (SO_2 ER **18.40%**, 95% CI 10.29–27.11) than post-pandemic (**2.91%**, 95% CI -4.82 to 11.28), while $PM_{2.5}$ and O_3, which lack a dominant indoor combustion source, showed no interaction ($p > 0.9$). The authors are careful: the formal interaction tests do not reach significance ($p = 0.13$, $p = 0.15$) and they label it hypothesis-generating. The mechanism they propose — confinement shifting exposure indoors toward residential heating — is precisely the failure mode of ambient-monitor exposure assignment, and it is unresolvable without personal or indoor measurement.
- **Source-diagnostic measurement keeps beating inventories.** Chen et al. constrain modelled biomass-burning black carbon in southern China with year-round ^{14}C at three sites and recover a normalised mean bias of **−68% to +28%** — an order-of-magnitude spread in a quantity inventories report as a number. Transboundary Southeast Asian contribution via free-tropospheric transport persists even in winter, when continental air masses dominate and are assumed to exclude it.
- **Emission control is redistributing nitrogen rather than removing it.** Across China IV to VI light-duty diesel, NO falls from >80% of reactive nitrogen while NH_3 (SCR slip) and N_2O (ammonia-slip catalyst) rise. Any network measuring NO_x alone is now blind to a growing fraction of what the fleet emits — a direct argument for speciated rather than surrogate measurement in traffic-adjacent deployments.
- **Two exposure pathways that ambient monitoring does not see.** Indoor air and dust supply up to **20% of the EFSA tolerable weekly intake** of PFAS for Faroese 15-year-olds, and an urban synthetic-turf field yields fine, submicron and nano-sized Fe/Si/Al/Br particles with Ni, Ag and Rh recovered from stub samplers worn *on the players* — microenvironmental sampling detecting what a fixed 24-hour $PM_{2.5}$ mean of 10.93 $\mu g/m^3$ reports as unremarkable.
- **Six of 26 records ship without an abstract.** All six are Elsevier, Springer or RSC DOIs absent from Crossref, Europe PMC and Semantic Scholar at build time. This is the fourth consecutive issue affected and the count is rising, not falling.

Corpus analytics

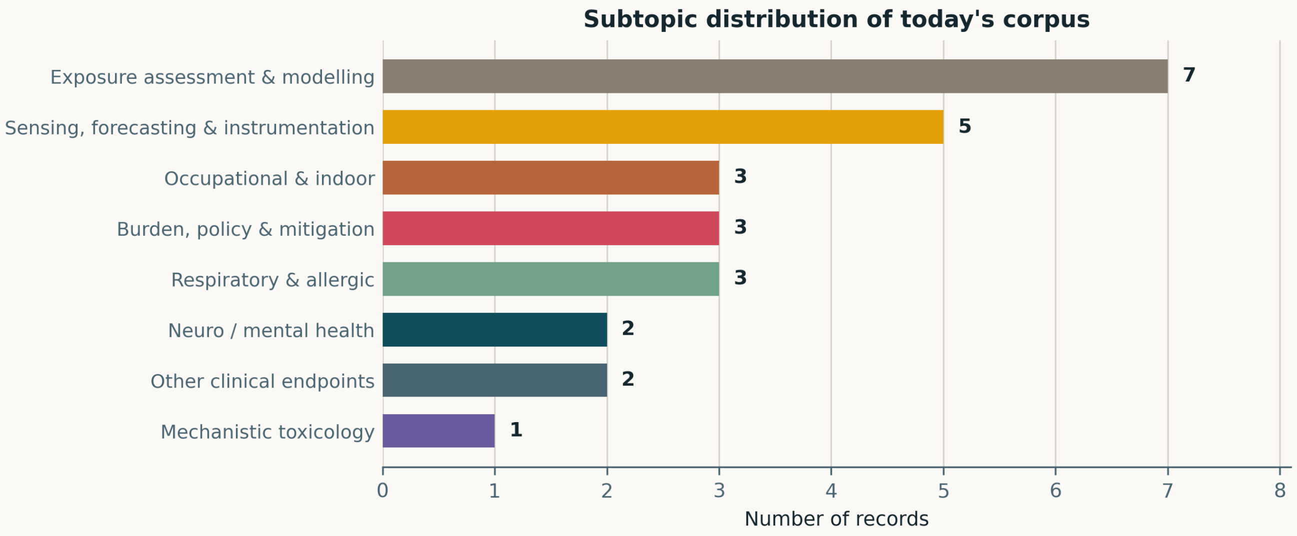


Fig. Subtopic assignment of the 26 carried records; single-label by dominant contribution. **Exposure assessment & modelling leads at 7**, ahead of **Sensing, forecasting & instrumentation at 5** — together 12 of 26, or 46% of the issue on the measurement side of the field. Occupational & indoor, Burden/policy and Respiratory & allergic tie at 3 each; Neuro and Other clinical endpoints at 2 each; Mechanistic toxicology at 1. **Two of the ten standing subtopics are unrepresented** — Cardiovascular & metabolic and Reproductive & developmental — both of which carried records as recently as last week.

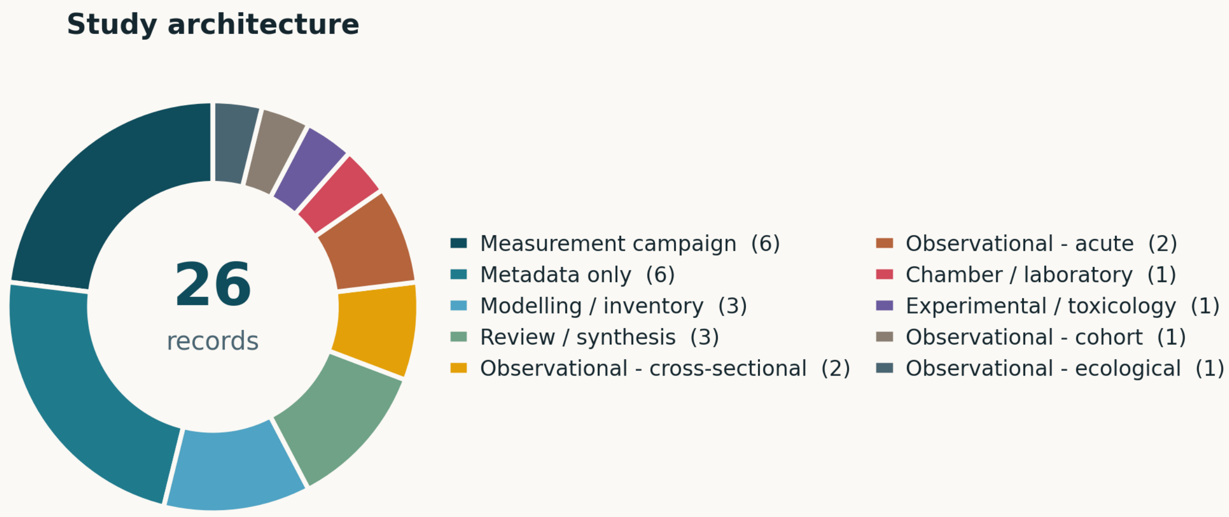


Fig. Study architecture across the 26 records. **Measurement campaign and Metadata only are tied at 6 each** — that is, as many records were carried without a retrievable abstract as were carried on the strength of a field campaign, which is the single most uncomfortable fact about this issue. Modelling / inventory and Review / synthesis follow at 3 each, then Observational cross-sectional and Observational acute at 2 each, with Chamber / laboratory, Experimental / toxicology, Observational cohort and Observational ecological at 1 apiece. *Note:* the combined f2 panel is not used this issue — with 14 distinct endpoint labels its y-axis text overlaps the donut centre and legend. The two panels are shown separately until that layout defect is fixed.

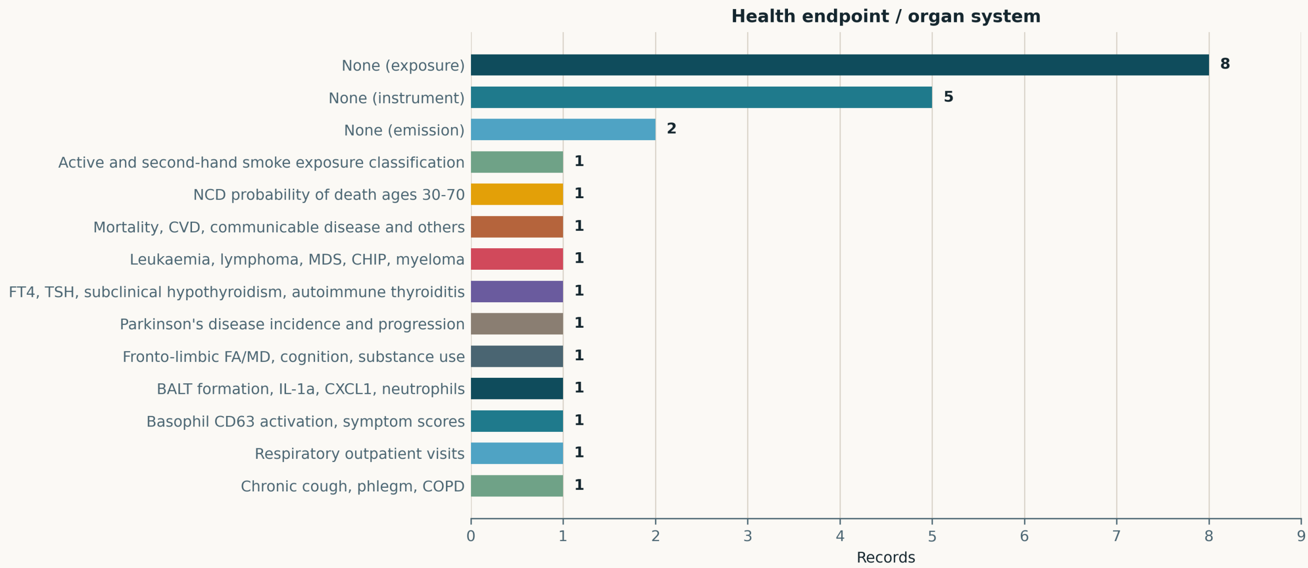
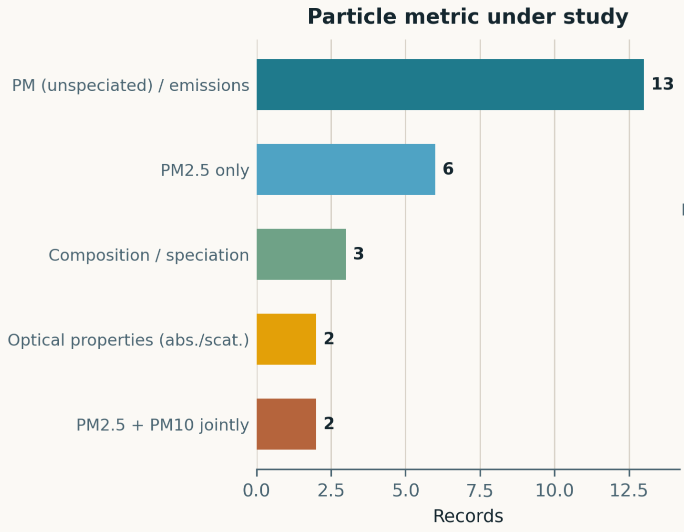


Fig. Health endpoint or organ system. **15 of 26 records report no health endpoint at all** — 8 exposure-only, 5 instrument-only, 2 emission-only — which is the structural signature of an instrumentation-weighted day. The remaining 11 records carry **11 distinct endpoints, each appearing exactly once**: thyroid function, BALT and IL-1α/CXCL1, fronto-limbic white matter, basophil CD63, respiratory outpatient visits, chronic cough/phlegm/COPD, haematologic malignancy, Parkinson's disease, NCD mortality, second-hand smoke classification, and a mortality/CVD/communicable-disease mapping review. No endpoint is replicated today, so nothing in this issue can corroborate anything else in it.



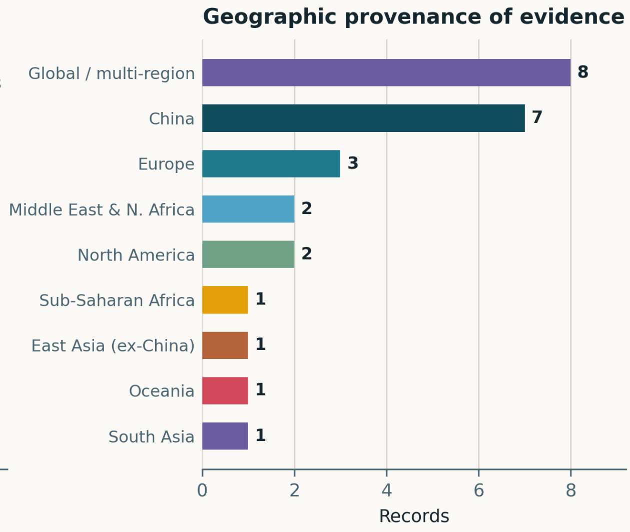


Fig. Left — particle metric. **The fallback bin "PM (unspeciated) / emissions" is still the largest at 13 of 26**, half the corpus, despite the four instrument-side classes added on 10 August. It is doing honest work here, because 6 of those 13 are metadata-only records whose particle metric genuinely cannot be read, but the bin remains the weakest column in the taxonomy. $PM_{2.5}$ only 6; composition / speciation 3; optical properties 2; $PM_{2.5}$ and PM_{10} jointly 2. Right — geographic provenance. **China 7 is the largest identifiable region**; "Global / multi-region" at 8 is inflated by the six metadata-only records and the two reviews, none of which state a site. Europe 3; Middle East & N. Africa and North America 2 each; South Asia, East Asia ex-China, Sub-Saharan Africa and Oceania 1 each. Three geo labels fell through the mapper on the first build and were pooled into the global bin; they are now mapped (see provenance).

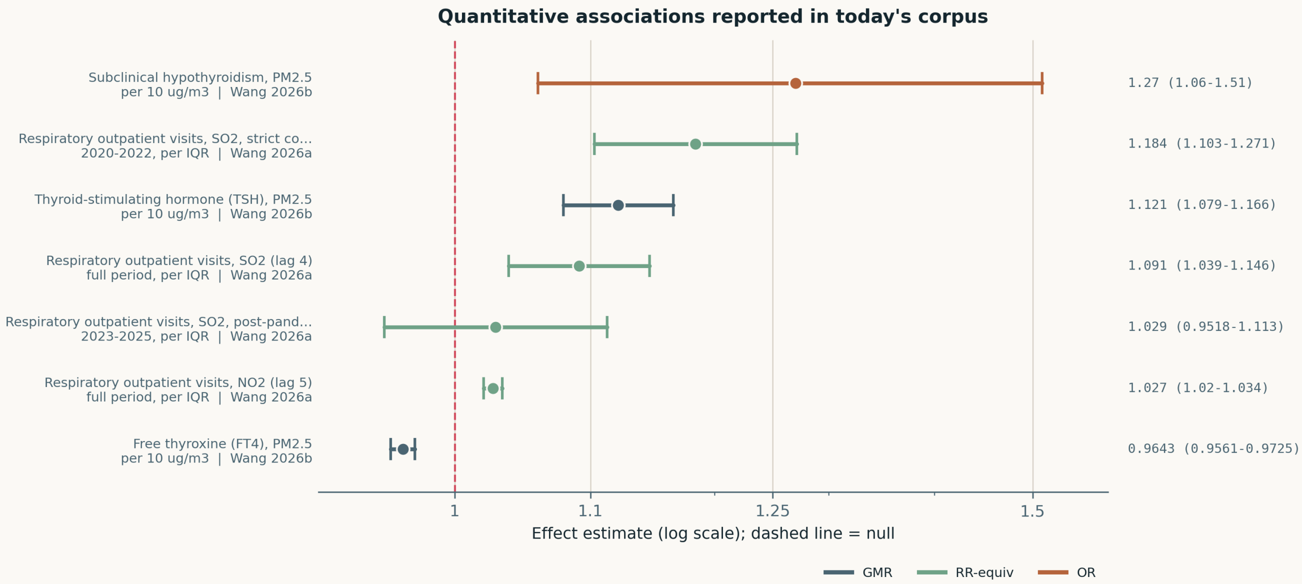


Fig. The seven interval estimates in today's corpus, log scale, dashed line at the null. **Six of seven exclude 1.0**; the exception is the post-pandemic SO_2 estimate, 1.029 (0.952–1.113), whose width is the whole point of that paper's phase comparison. Metrics are heterogeneous by design and are colour-coded: OR for subclinical hypothyroidism, GMR (geometric-mean ratio) for the two continuous thyroid biomarkers, and RR-equiv for the four excess-risk percentages, converted as $1 + ER/100$ so they sit on a common ratio axis. **They are not poolable** — two studies, two populations, two exposure definitions — and no summary estimate is drawn.

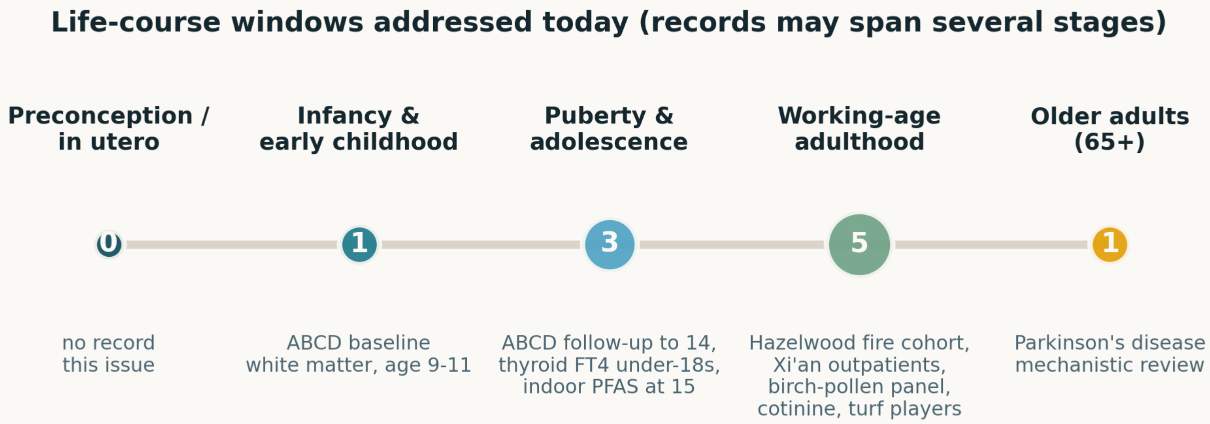


Fig. Life-course windows addressed. **Working-age adulthood dominates at 5**, against 3 in puberty/adolescence, 1 in late childhood, 1 in older adults and **zero in utero** — the first issue since 9 August with no preconception or prenatal record. *This figure was a defect until this issue*: `plots.py` carried a silent hardcoded fallback that rendered counts and exemplars from unrelated earlier issues whenever the corpus JSON lacked a `LIFECOURSE` key. It is now a hard build failure, and the counts above are read off today's records.

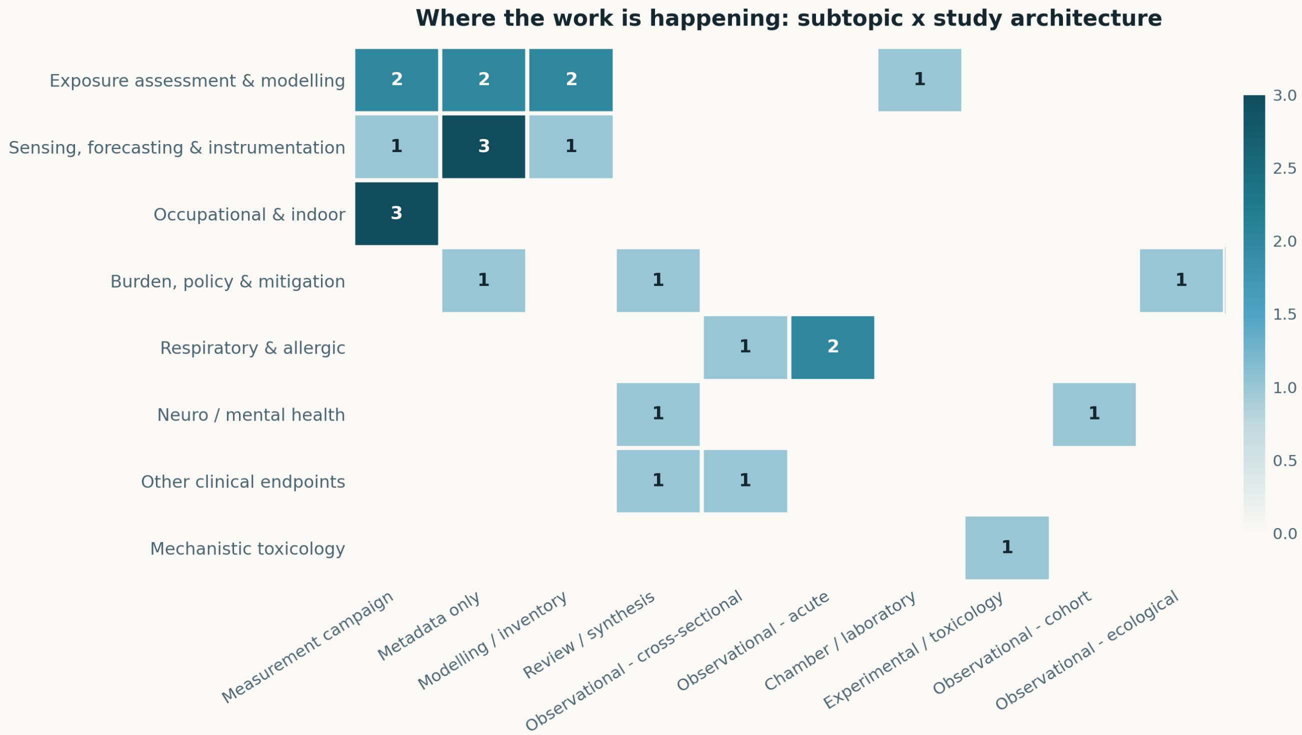


Fig. Subtopic × study architecture; blank cells are true zeros. The two densest cells are **Sensing** × **Metadata only** (**3**) and **Occupational & indoor** × **Measurement campaign** (**3**). The first is the diagnostic one: the subtopic this watch exists to track is also the subtopic least likely to arrive with a readable abstract, because it publishes in Elsevier and Springer atmospheric-science titles that deposit metadata without abstracts. Exposure assessment spreads evenly across measurement, metadata-only and modelling (2 each) plus one review. Respiratory & allergic is the only health subtopic with more than one architecture at $n > 1$ (2 acute, 1 cross-sectional).

Paper-level digest

Records are grouped by subtopic and ordered within group by first author. Tier A/B/C is a judgement about how much weight the design can carry, not about the journal. Records marked METADATA ONLY had no abstract in Crossref, Europe PMC, Semantic Scholar or Scholar Gateway at build time and are listed rather than summarised.

Sensing, forecasting & instrumentation — 5 records

Balasubramanian et al. 2026 Environ Pollut

Calibration model for PurpleAir-monitored $PM_{2.5}$ across the United States

Random-forest correction for PurpleAir $PM_{2.5}$ trained on 2019–2021 and evaluated on a held-out 2022, using only sensor-reported inputs — PM_1, $PM_{2.5}$, PM_{10}, temperature, pressure, relative humidity — with no external meteorology. Pearson r against co-located EPA monitors is **0.90–0.96** across buffers from 500 m to 5 km; multiple calibrated products are released at different buffer radii. Deliberately interpretable rather than a deep network, and explicitly aimed at regions without regulatory infrastructure. *The limitation is the metric*: r measures covariation, not agreement, and is blind to the multiplicative bias hygroscopic growth actually produces. Without RMSE, slope or bias the accuracy claim is unverified. Still the most directly transferable record of the day for a distributed PM network — a national, sensor-only, temporally validated correction is the exact artefact a deployment needs.

doi: 10.1016/j.envpol.2026.128935

Albashir 2026 Res Sq (preprint)

A physics-informed digital twin framework for health vigilance and indoor air quality decision support

Preprint proposing a physics-informed digital twin for indoor air quality in healthcare-oriented environments, arguing that monitoring-only systems do not support proactive decisions. **Not peer reviewed, single author, no deployment, no validation against measured concentrations reported in the abstract.** Carried only because the architectural claim — coupling a physical model of a room to a sensor stream so the system can extrapolate beyond sensor locations — is the same idea as physics-informed calibration on the ambient side, and the indoor literature is where it is being tried first. Treat as a pointer to a design pattern, not as evidence. Relevance to distributed sensing: physics-informed correction is the leading candidate for generalising calibration away from co-location sites, and this is a low-cost read on how that framing is developing indoors.

doi: 10.21203/rs.3.rs-10034990/v1

***Atmos Environ* 2026a METADATA ONLY** Atmos Environ

Reliability-constrained identification of primary and elevated aerosol structures from a single tilted Mie lidar under dust-influenced conditions

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Listed, not summarised. Carried because the title states a specific and testable capability: retrieving both the primary and the elevated aerosol layer from a *single tilted* Mie lidar, with an explicit reliability constraint, under dust loading that normally defeats single-wavelength inversion. Single-instrument retrieval of layer structure is the difference between a research lidar and one a network can afford, and dust is the condition under which the retrieval usually fails. Whether the paper delivers that is unknown until an abstract appears.

doi: 10.1016/j.atmosenv.2026.122297

***Atmos Environ* 2026b METADATA ONLY** Atmos Environ

Stable carbon isotopic composition of PM_1 emissions from biomass and coal combustion under monitored combustion conditions

METADATA ONLY — no abstract retrievable at build time. Carried because $\delta^{13}C$ signatures for PM_1 measured under *monitored* combustion conditions are the reference values that source-apportionment studies need and usually borrow from decades-old literature. Biomass and coal are the two sources whose separation matters most in residential-heating regions, and combustion-condition control addresses the standing objection that isotopic signatures drift with burn phase. If the reported ranges are tight this is a reference dataset; if they overlap it undermines $\delta^{13}C$ apportionment. Unresolvable without the text.

doi: 10.1016/j.atmosenv.2026.122296

***Aerosol Air Qual Res* 2026 METADATA ONLY** Aerosol Air Qual Res

Identifying winter $PM_{2.5}$ sources in an urban environment using receptor modeling and nonparametric trajectory analysis

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Carried because the pairing of receptor modelling with *nonparametric* trajectory analysis is methodologically specific: it drops the functional-form assumption that conventional trajectory statistics impose on the source–receptor relationship, which is the assumption most likely to fail in complex winter meteorology with shallow boundary layers. Site, sample size and factor count are all unstated. Listed for the register, not weighted.

doi: 10.1007/s44408-026-00157-8

Exposure assessment & modelling — 7 records

Chen et al. 2026 Environ Sci Technol

Inventory uncertainty and transboundary transport of biomass-burning black carbon in southern China

Year-round radiocarbon measurement of biomass-burning black carbon at two regional and one background site in southern China, compared against tailored atmospheric-transport simulations. The model reproduces seasonality but the normalised mean bias spans **-68% to $+28\%$** — a factor-of-two-plus spread in a quantity inventories report without error bars, attributed largely to unmonitored biomass burning in mainland China. Residential burning across East Asia dominates; Southeast Asian transport matters in summer and, via free-tropospheric pathways, **persists in winter** when continental air masses are assumed to exclude it. *Limitation*: three sites cannot constrain a regional inventory, and ^{14}C apportionment inherits uncertainty from the assumed fossil/non-fossil end-members. Relevance: this is the cleanest current demonstration that source-diagnostic *measurement*, not finer model resolution, is what bounds inventory error — the argument for putting source-resolving capability into distributed networks.

doi: 10.1021/acs.est.6c00449

Deng et al. 2026 Atmos Chem Phys

Impact of black carbon on daytime valley and slope winds in idealised simulations

High-resolution idealised WRF-Chem experiments, run with and without radiatively interactive black carbon, on thermally driven valley circulation. Absorption warms the upper boundary layer and cools lower levels, increasing stability, reducing surface heating, and leaving up-slope winds **weaker and confined to a shallower layer**. The feedback is self-reinforcing: the circulation that ventilates a valley is suppressed by the aerosol it would otherwise disperse. *Limitation*: two idealised simulations with prescribed BC, no observational constraint, no real terrain — the sign of the effect is credible, the magnitude is not transferable. Relevance to monitoring: in mountain basins the relationship between emissions and measured concentration is not linear, and siting a network to sample a circulation that the pollution itself modifies requires the vertical dimension, not a denser surface grid.

doi: 10.5194/acp-26-11171-2026

***Environ Sci Atmos* 2026a** Environ Sci Atmos

Multi-scale variability and lagged meteorological controls of $PM_{2.5}$ in Delhi

Wavelet coherence and causality analysis of Delhi $PM_{2.5}$ against meteorological drivers, decomposing the record by timescale rather than fitting a single regression. **The abstract is truncated at source** — it stops mid-sentence after stating the aim, and no coherence bands, lag lengths, data period or station count are recoverable. What can be said is that the method is the right one for the problem: Delhi's $PM_{2.5}$ is driven by processes on separable scales (diurnal boundary-layer collapse, synoptic ventilation, seasonal crop-residue burning) that a time-domain model conflates, and lagged causality is what forecasting a distributed network needs. Carried for the register with its numbers unstated. Re-check on the weekly abstract sweep.

doi: 10.1039/d6ea00071a

Li et al. 2026a Environ Sci Technol

Upgrading diesel vehicle after-treatment technology significantly changed the composition of reactive nitrogen in exhaust

Chassis-dynamometer and controlled testing of light-duty diesel vehicles across China IV, V and VI standards. Nitrogen speciation shifts fundamentally: **NO exceeds 80% of reactive nitrogen under China IV**, while NH_3 and N_2O become prominent under China VI. Mechanisms are attributed component-wise — diesel oxidation catalysts raise the NO_2/NO_x ratio, SCR produces NH_3 slip, ammonia-slip catalysts generate N_2O. SHAP analysis of tree models identifies emission standard, exhaust temperature and vehicle speed as dominant drivers. *Limitation*: no fleet sample size or vehicle count is given, and dynamometer cycles under-represent cold starts and real-world transients where SCR is least effective. Relevance: NH_3 is a secondary-PM precursor and N_2O a greenhouse gas, so successive NO_x controls have moved the problem rather than closed it. A traffic-adjacent network measuring NO_x alone now under-reports the emission it is meant to characterise.

doi: 10.1021/acs.est.6c00994

Lin et al. 2026 Environ Sci Technol

Three-millennia alpine-lake records of divergent atmospheric Cd, Pb, Sb and Hg pollution in southwestern China

Radiometrically dated sediment cores from three alpine lakes at ∼4000 m in Yunnan reconstruct 3000 years of atmospheric Sb, Pb, Cd and Hg deposition. Onset is metallurgy-specific and spatially staggered: Sb and Pb from the mid-1400s in northeastern Yunnan, ∼200 years later in the west with Cu/Ag smelting; Cd only after the early 1700s with Zn and Cu production. Preindustrial peaks fall between the early 1800s and 1900 CE. **Since the 1950s Cd has reached unprecedented levels while Sb and Pb remain below their preindustrial maxima**, and Hg rises monotonically from the mid-1400s. *Limitation*: lake sediment integrates catchment erosion as well as atmospheric deposition, and the separation depends on normalisation assumptions not visible in the abstract. Relevance: a 3000-year baseline is the only way to know what "background" means for a metal, and metal-resolved PM is where low-cost sensing has no capability at all.

doi: 10.1021/acs.est.5c17960

***Atmos Environ* 2026c METADATA ONLY** Atmos Environ

Source-specific black carbon responses to short-term interventions and cross-year emission changes in coastal Qingdao

METADATA ONLY — no abstract retrievable. Carried because source-specific BC response to *short-term* interventions is a natural-experiment design, and coastal sites separate shipping from land-based combustion in a way inland sites cannot. Read alongside Chen et al. in this issue, which finds inventory-based BC attribution biased by −68% to +28%: an intervention study measures the response directly and sidesteps that. Sample period, apportionment method and intervention definition are all unstated.

doi: 10.1016/j.atmosenv.2026.122299

***Atmos Pollut Res* 2026 METADATA ONLY** Atmos Pollut Res

High-resolution CFD modelling of traffic-related air pollutant dispersion in an urban area of Istanbul

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar. Carried because building-resolving CFD is the reference against which cheap dispersion proxies — Gaussian plume, land-use regression, distance-to-road — are judged, and because street-canyon gradients at the scale CFD resolves are exactly what a dense low-cost network claims to capture and what a regulatory monitor cannot. Whether the model is validated against measurement, and at what resolution, is unknown. Listed, not weighted.

doi: 10.1016/j.apr.2026.103169

Occupational & indoor 3 records

Alqahtani et al. 2026 PLoS One

Biomarker-based assessment of tobacco smoke exposure in Saudi females via urinary cotinine

LC-MS/MS quantification of urinary cotinine in 133 Saudi women recruited May–December 2024, validated to ICH-M10 (linearity 2–500 ng/mL, $R^2 \geq 0.9992$, accuracy 90–107%, CV <10%), with a population-specific cut-off of **7 ng/mL**. Cotinine was detected in **48 of 133** samples although **88.7% self-reported as non-smokers**, and 45.9% reported household second-hand smoke. *Limitations*: $n = 133$ is small for establishing a population cut-off; the detected range (0.02–7.693 ng/mL) sits almost entirely *below* the proposed 7 ng/mL threshold, so the cut-off classifies nearly every detected sample as non-smoking and its derivation is not defensible from the reported distribution; and cotinine's short half-life indexes recent exposure only. Relevance: a clean demonstration that self-report and biomarker diverge sharply under social desirability pressure — the same argument for objective personal measurement over questionnaires that motivates wearable PM sensing.

doi: 10.1371/journal.pone.0355166

Herkert et al. 2026 Environ Sci Technol

Contribution of indoor air and dust to children's PFAS exposure in the Faroe Islands

Measurement of nonvolatile precursors, volatile neutral precursors and legacy ionic PFAS in air and dust across **40 Faroese homes**, with estimated daily intake computed for 15-year-olds. EDI ranges 0.001 / 0.012 / 0.129 ng/kg bw across low, intermediate and high scenarios, equal to roughly **0.2%, 2% and 20% of the EFSA tolerable weekly intake**, against ∼60% from pilot-whale consumption. Indirect exposure to perfluoroalkyl carboxylic acids via air dominates, followed by PFOS in dust. *Limitations*: scenario-based EDIs are not measured intakes, and the 40-home sample cannot support the tail of the distribution where the 20% figure sits. Relevance: the indoor compartment supplies a fifth of a regulatory reference dose in the high scenario for a population with no local source — a direct case for indoor sampling as a routine part of exposure assessment rather than a specialist add-on.

doi: 10.1021/acs.est.6c02003

Iavazzo et al. 2026 Environ Geochem Health

Particle characterization in an urban synthetic turf: turf and vehicular wear

XRF, SEM-EDS and dynamic light scattering on dust, turf fibre and crumb-rubber infill from an urban synthetic pitch, with stub-based sampling worn *on players*. Irregular and elongated fine, submicron and nano-sized particles were observed; Fe dominated by XRF, with Fe, Si, Al and Br-bearing particles by SEM-EDS and **Ni, Ag and Rh** recovered from the player-worn stubs — Rh being a clear three-way-catalyst tracer, so the pitch is receiving vehicular as well as turf-wear particles. Ambient $PM_{2.5}$ averaged **10.93** $\mu\mathbf{g/m^3}$ (24-h) and O_3 reached 41.12 ppb. *Limitations*: one field, no personal inhaled dose, and the deterministic USEPA assessment — which found no significant non-carcinogenic metal risk — is mass-based and cannot speak to nanoparticle toxicity. Relevance: the contrast between an unremarkable regulatory 24-h mean and a demonstrably loaded microenvironment is the clearest argument in this issue for personal and microenvironmental sampling.

doi: 10.1007/s10653-026-03407-1

Respiratory & allergic — 3 records

Bogunia-Kubik et al. 2026 Allergy

Influence of environmental pollution on basophil reactivity to birch pollen allergens

Fifty birch-allergic patients and 25 non-atopic controls followed over the 2023 and 2024 pollen seasons in Kraków, with app-recorded daily nasal, ocular and asthma symptoms against continuously monitored $PM_{2.5}$, PM_{10}, O_3 and birch pollen. Basophil activation tests used commercial extract, recombinant Bet v 1, and **natural pollen collected from highly and less polluted sites**. Pollen from polluted sites induced significantly stronger CD63 expression ($p < 0.0001$), and asthma symptoms correlated with both pollen and PM ($p < 0.05$, mainly 2023). *Limitations*: the central comparison is site-level, so pollution is confounded with everything else that differs between the two collection sites; $n = 75$ over two seasons with a season-specific correlation is thin; and CD63 is an ex-vivo surrogate. Relevance: if pollution modifies allergen *potency* rather than only co-occurring with it, exposure models that treat pollen and PM as independent additive terms are mis-specified.

doi: 10.1111/all.70474

Govindaraju et al. 2026 J Occup Environ Med

Protective role of dietary antioxidant intake on long-term effects of extreme $PM_{2.5}$ exposure on respiratory health

Cross-sectional analysis of 448 Hazelwood Health Study adults, with individual $PM_{2.5}$ from the 2014 Morwell coal-mine fire reconstructed by combining modelled fire-related concentrations with contemporaneous time–location diaries — a better exposure assignment than most disaster epidemiology achieves. Vitamins A and E, magnesium and zinc attenuated the $PM_{2.5}$–chronic-cough association, and all but magnesium the phlegm association; omega-3 attenuated the $PM_{2.5}$–COPD association. *Limitations are severe*: diet was measured cross-sectionally years after a 2014 exposure, so reverse causation and healthy-diet confounding are unaddressed; **no effect sizes or intervals are reported in the abstract**; and multiple testing across several nutrients and several outcomes is not accounted for. Relevance: effect-modification analyses of this kind demand exposure precision that only time-resolved personal or time-activity-weighted measurement supplies.

doi: 10.1097/JOM.0000000000003856

Wang et al. 2026a Front Public Health

COVID-19 social interventions amplified indoor-source pollutant–respiratory associations in Xi'an

Six-year time series of **2,052,777** respiratory outpatient visits over 2192 days at a tertiary hospital in Xi'an (2020–2025), stratified into strict control, transition and post-pandemic phases, with quasi-Poisson GAMs and pollutant-by-phase interaction terms. Full-period associations were strongest for SO_2 (lag 4, ER **9.11%**, 95% CI 3.86–14.64) and NO_2 (lag 5, **2.72%**, 2.05–3.40). Phase-stratified, SO_2 was **18.40% (10.29–27.11)** under strict control versus **2.91% (−4.82 to 11.28)** post-pandemic; $PM_{2.5}$ and O_3 showed no interaction ($p > 0.9$). *The authors state plainly that the interaction tests do not reach significance* ($p = 0.13$, $p = 0.15$), and single-hospital outpatient counts confound exposure effects with healthcare-seeking behaviour that the pandemic also changed. Relevance: the proposed indoor-amplification mechanism is unfalsifiable with ambient monitors alone, which is the argument for indoor and personal measurement in time-series designs.

doi: 10.3389/fpubh.2026.1896806

Neuro / mental health — 2 records

Sullivan et al. 2026 Dev Psychopathol

Longitudinal impacts of childhood neighborhood air pollution on white matter integrity and cognition

ABCD Study, $n = 4{,}511$ children (47.4% female, baseline mean age 9.9), with baseline $PM_{2.5}$ tested against fronto-limbic fractional anisotropy and mean diffusivity across follow-up, controlling for neighbourhood factors. **The direction of association reverses with age**: at 9, higher $PM_{2.5}$ predicted *increased* FA in anterior thalamic radiation, fornix and uncinate; by 14, *reduced* FA in all tracts. In superior longitudinal fasciculus, higher MD at 9 and reduced FA at 12 and 14. **Associations with cognition and substance use were null.** *Limitations*: baseline-only exposure ignores subsequent trajectory; neighbourhood-level assignment carries substantial misclassification; and a sign reversal is equally consistent with accelerated early myelination followed by decline, or with residual confounding that changes with age. Relevance: a design that needs longitudinal, individually resolved exposure and had only a single neighbourhood-level baseline value — the precise gap distributed personal sensing addresses.

doi: 10.1017/S095457942610159X

***Parkinsons Dis* review 2026** Parkinsons Dis

Air pollution and Parkinson's disease pathology: clinical evidence and molecular mechanisms

Narrative review linking long-term NO_x, PM and O_3 exposure to Parkinson's incidence and to worse prognosis, with proposed mechanisms spanning gut-microbiome alteration, oxidative stress, neuroinflammation and promotion of α-synuclein aggregation. *No pooled estimate, no systematic search strategy, no risk-of-bias assessment and no quantitative synthesis* — it is a mechanistic orientation piece, not evidence. Tier C accordingly. The gut-microbiome route is the one worth noting, because it implies an ingestion and mucosal pathway for deposited particles that inhaled-dose models do not represent, and because it predicts that the relevant exposure metric is cumulative deposited mass in a specific size range rather than ambient concentration. Carried for the register and for that single mechanistic pointer.

doi: 10.1155/padi/6914112

Other clinical endpoints 2 records

***Curr Hematol Malig Rep* 2026** Curr Hematol Malig Rep

Linking air pollution and hematologic malignancies: from epidemiology to molecular mechanisms

Review of outdoor air pollution and haematologic malignancy across acute and chronic leukaemias, lymphomas, MDS, CHIP and myeloma, implicating PM, benzene, NO_2, SO_2, arsenic and ethylene oxide through oxidative stress, DNA damage, epigenetic dysregulation, chronic inflammation and haematopoietic stem-cell dysfunction. Evidence is graded strongest for AML with benzene; ALL associations with traffic-related pollution, NO_2 and PM are described as less consistent; evidence for chronic leukaemias and myeloma is limited. *Not systematic*: no search protocol, no pooled estimates, no bias assessment, and the authors themselves flag exposure misclassification and residual confounding. The CHIP link is the forward-looking one — a clonal intermediate would give this literature a measurable early endpoint. Relevance: benzene and PM co-vary in traffic sources, and separating them requires speciated measurement that mass monitors cannot provide.

doi: 10.1007/s11899-026-00785-2

Wang et al. 2026b J Hazard Mater

Ambient $PM_{2.5}$ and thyroid health: 24,333 participants in Zhejiang, China

Community-based cross-section of **24,333** participants (2022–2025) spanning minors and adults, with $PM_{2.5}$ assigned from geocoded residential addresses and mixed models estimating associations with FT4, TSH, subclinical hypothyroidism and autoimmune thyroiditis. Per 10 $\mu g/m^3$: FT4 $-$**3.57% (95% CI $-$4.39, $-$2.75)**, TSH +**12.15% (7.92, 16.56)**, subclinical hypothyroidism OR **1.27 (1.06, 1.51)**; no clear association with autoimmune thyroiditis. The FT4 effect was larger in under-18s and in BMI <18.5; the TSH and hypothyroidism effects larger in those $\geq$50 and BMI $\geq$24. *Limitations*: cross-sectional, so exposure precedence is assumed not observed; residential-address assignment ignores time–activity; and iodine status is invoked as a rationale but its adjustment is not described in the abstract. The internally coherent FT4-down/TSH-up pattern argues against pure chance. Relevance: the paediatric and low-BMI susceptibility windows require exposure resolved at the individual rather than the address.

doi: 10.1016/j.jhazmat.2026.143231

Mechanistic toxicology 1 record

Ichinose et al. 2026 J Hazard Mater

Time-dependent BALT formation and pulmonary inflammation induced by different particulate matter types

Murine time-course comparing TiO_2 nanoparticles, diesel exhaust particles and Asian sand dust, with and without ovalbumin. All three supported bronchus-associated lymphoid tissue formation, more strongly with allergen. With allergen, an early IL-1α rise preceded type-2 responses and BALT expansion; **without allergen, BALT still formed via CXCL1 and neutrophil accumulation**, most clearly after repeated TiO_2 — two distinct routes to the same structure. 3D reconstruction showed tertiary lymphoid structures extending from large airways distally, and TiO_2 was localised in alveoli, alveolar macrophages, BALT and hilar lymph nodes. *Limitations*: instilled or high-dose murine exposure is not ambient inhalation; no dose–response to ambient concentrations; and the three particle types differ in dose, size and surface chemistry simultaneously. Relevance: particle *identity*, not mass, determines which inflammatory route is taken — the standing mechanistic case for composition-resolved monitoring.

doi: 10.1016/j.jhazmat.2026.143239

Burden, policy & mitigation 3 records

Lee et al. 2026 J Korean Med Sci

Toward a unified framework for climate and health research: gaps in exposure metrics and health outcomes

Mapping review of **611** original human studies (PubMed, Cochrane, Embase; January 2000 – June 2024) on climate-related exposures and health. The distribution is sharply skewed: **526 studies on temperature and temperature-related hazards, 88 on precipitation, and only 39 on air pollution**, with extreme weather barely represented. Outcomes concentrate in all-cause mortality (249), cardiovascular disease (159) and communicable disease (148), while neurodegenerative disease (8) and endocrine disorders (4) are almost absent. Exposure-metric definitions are described as substantially heterogeneous. *Limitations*: counts are of studies, not of evidence quality, and English-only inclusion biases the geography. Relevance: the two endpoints this issue adds — thyroid function and fronto-limbic white matter — sit in the two thinnest outcome categories the review identifies, and the metric heterogeneity it documents is the same problem that makes cross-study pooling of PM exposure estimates unreliable.

doi: 10.3346/jkms.2026.41.e294

Okeke et al. 2026 Environ Monit Assess

Agricultural methane emissions and non-communicable disease mortality in West Africa

Ecological panel of 11 West African countries, 2015–2020, relating agricultural methane (Mt CO_2e, World Bank WDI) to the probability of dying from NCDs between ages 30 and 70, using pooled OLS, country fixed effects and mixed models, with The Gambia ($-$28.3% emissions) as a descriptive contrast. *This is a weak design and should be read as such*: 11 countries $\times$ 6 years is 66 country-years, the exposure is a national accounting aggregate with no measured concentration and no plausible direct mechanism to NCD mortality, and the outcome is a modelled WHO indicator rather than observed deaths. Ecological fallacy and confounding by development trajectory are unaddressed. Tier C, carried only because it is one of the few Sub-Saharan records in this watch and marks where the climate–health literature is expanding without measurement infrastructure — which is itself the finding.

doi: 10.1007/s10661-026-15800-5

***Atmos Environ* 2026d METADATA ONLY** Atmos Environ

Drivers of air pollutant and greenhouse gas emissions since 2010 in the Greater Bay Area, China

METADATA ONLY — no abstract retrievable at build time. Carried for the register because decomposition of co-emitted air pollutants and greenhouse gases over a 15-year window in a single megacity cluster is the analysis that determines whether observed PM improvement came from control technology, fuel switching, activity shift or economic structure — and those four have different implications for what happens next. Method, pollutant coverage and decomposition scheme are all unstated. Lowest-weight record in the issue.

doi: 10.1016/j.atmosenv.2026.122301

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Parkinsons Dis review	Air pollution and Parkinson's disease pathology: clinical evidence and molecular mechanisms	*Parkinsons Dis*	Mechanistic review	PM, NOx, O3	Global
Sullivan et al.	Longitudinal impacts of childhood neighborhood air pollution exposure on white matter integrity and cognition	*Dev Psychopathol*	Prospective cohort	$PM_{2.5}$	United States
Respiratory & allergic					
Bogunia-Kubik et al.	Influence of environmental pollution on basophil reactivity to birch pollen allergens in atopic and non-atopic subjects	*Allergy*	Panel study	$PM_{2.5}$, PM_{10} (and O3)	Poland
Govindaraju et al.	Protective role of dietary antioxidant intake on long-term effects of extreme $PM_{2.5}$ exposure on respiratory health	*J Occup Environ Med*	Cross-sectional (cohort baseline)	Fire-sourced $PM_{2.5}$	Australia
Wang et al.	COVID-19 social interventions were associated with amplified, not attenuated, indoor-source pollutant-respiratory associations in Xi'an	*Front Public Health*	Retrospective time series	$PM_{2.5}$, SO2, NO2, O3	China
Mechanistic toxicology					
Ichinose et al.	Time-dependent bronchus-associated lymphoid tissue formation and pulmonary inflammation induced by different particulate matter types	*J Hazard Mater*	Animal model	TiO2 nanoparticles, diesel exhaust particles, Asian sand dust	Japan
Occupational & indoor					
Alqahtani et al.	Biomarker-based assessment of tobacco smoke exposure in Saudi females via urinary cotinine	*PLoS One*	Repeated-measures biomarker	Tobacco smoke (urinary cotinine)	Saudi Arabia
Herkert et al.	Contribution of indoor air and dust to children's PFAS exposure in the Faroe Islands	*Environ Sci Technol*	Field measurement	Indoor air and settled dust, PFAS	Faroe Islands
Iavazzo et al.	Particle characterization in an urban synthetic turf: insights into turf and vehicular wear	*Environ Geochem Health*	Field measurement	Fine, submicron and nano-sized particles; Fe, Si, Al, Br, Ni, Ag, Rh	Italy
Sensing, forecasting & instrumentation					
Aerosol Air Qual Res	Identifying winter $PM_{2.5}$ sources in an urban environment using receptor modeling and nonparametric trajectory analysis	*Aerosol Air Qual Res*	Metadata only (no abstract)	$PM_{2.5}$, source-resolved	Not stated
Albashir 2026	A physics-informed digital twin framework for health vigilance and indoor air quality decision support	*Res Sq* (*preprint*)	Simulation (system-level)	Indoor PM and IAQ composite	Not stated
Atmos Environ	Reliability-constrained identification of primary and elevated aerosol structures from a single tilted Mie lidar under dust-influenced conditions	*Atmos Environ*	Metadata only (no abstract)	Backscatter / aerosol layer structure	Not stated
Atmos Environ	Stable carbon isotopic composition of PM1 emissions from biomass and coal combustion under monitored combustion conditions	*Atmos Environ*	Metadata only (no abstract)	PM1, stable carbon isotopes	Not stated
Balasubramanian et al.	Calibration model for PurpleAir-monitored $PM_{2.5}$ across the United States	*Environ Pollut*	Calibration model comparison	PM1, $PM_{2.5}$, PM_{10}	United States
Exposure assessment & modelling					
Atmos Environ	Source-specific black carbon responses to short-term interventions and cross-year emission changes in coastal Qingdao	*Atmos Environ*	Metadata only (no abstract)	Black carbon, source-resolved	China
Atmos Pollut Res	High-resolution CFD modelling of traffic-related air pollutant dispersion in an urban area of Istanbul	*Atmos Pollut Res*	Metadata only (no abstract)	Traffic-related PM and NOx	Turkiye
Chen et al.	Inventory uncertainty and transboundary transport of biomass-burning black carbon in southern China from radiocarbon plus modelling	*Environ Sci Technol*	CTM simulation + source attribution	Black carbon, 14C source-resolved	China
Deng et al.	Impact of black carbon on daytime valley and slope winds in idealised WRF-Chem simulations	*Atmos Chem Phys*	Theory / simulation	Black carbon, radiative absorption	Idealised terrain
Environ Sci Atmos	Multi-scale variability and lagged meteorological controls of $PM_{2.5}$ in Delhi from wavelet coherence and causality analysis	*Environ Sci Atmos*	Monitoring-record analysis	$PM_{2.5}$	India

First author	Focus	Journal	Design	Metric	Region
Li et al.	Upgrading diesel vehicle after-treatment technology significantly changed the composition of reactive nitrogen in exhaust	*Environ Sci Technol*	Vehicle emission experiment	Reactive nitrogen speciation (NO, NO2, NH3, N2O)	China
Lin et al.	Three-millennia alpine-lake records of divergent atmospheric Cd, Pb, Sb and Hg pollution in southwestern China	*Environ Sci Technol*	Environmental surveillance campaign	Atmospheric metal deposition: Cd, Pb, Sb, Hg	China
Burden, policy & mitigation					
Atmos Environ	Drivers of air pollutant and greenhouse gas emissions since 2010 in the Greater Bay Area, China	*Atmos Environ*	Metadata only (no abstract)	PM and co-emitted pollutants, emissions	China
Lee et al.	Toward a unified framework for climate and health research: gaps in exposure metrics and health outcomes	*J Korean Med Sci*	Mapping review	Climate exposure metrics incl. air pollution	Global
Okeke et al.	Agricultural methane emissions and non-communicable disease mortality in West Africa, 2015-2020	*Environ Monit Assess*	Ecological panel, fixed effects	Agricultural CH4 (Mt CO2e)	West Africa
Other clinical endpoints					
Curr Hematol Malig Rep	Linking air pollution and hematologic malignancies: from epidemiology to molecular mechanisms	*Curr Hematol Malig Rep*	Mechanistic review	PM, benzene, NO2, SO2, arsenic, ethylene oxide	Global
Wang et al.	Ambient $PM_{2.5}$ and thyroid health: a cross-sectional study of 24,333 participants in Zhejiang, China	*J Hazard Mater*	Cross-sectional (multistage)	$PM_{2.5}$	China

Provenance and method

Entry window **2026-08-11 to 2026-08-11**, contiguous with the 10 August issue; no gap. **PubMed** connector, three independent queries against `[Date - Entry]`: health axis 9, instrumentation axis 10, broad MeSH sweep 23 — **33 distinct** records, **18 carried**. Ten of the 33 carry an Entrez create date of 10 August but an `[EDAT]` of 11 August; none was in `seen.json`, so they belong to this issue and the two date fields are not interchangeable. Local harvester: PubMed health 18, PubMed sensing 1, Europe PMC `CREATION_DATE` 1 (**1 carried**, a Research Square preprint), Crossref by ISSN across 17 journals 12 (**7 carried**). **Consensus** returned 20, top 3 retrievable on the free tier, **0 carried** — all three resolved to Crossref creation dates of 26 March 2026, 11 November 2025 and 21 April 2026 and all three were already in `seen.json` on DOI. That is the **fifth consecutive run** confirming Consensus as a relevance index that cannot resolve a single day; it continues to earn its place as a duplicate check, not as a recency leg. **arXiv** 60 records, **1** published on or after 5 August and off-topic (a climate–health bibliometric survey). **OpenAlex** HTTP 429, **ninth consecutive run**. **ClinicalTrials.gov** returned **0** trials registered or updated on 11 August; `trials.json` unchanged, and the trial watch is surfaced in the weekly rollup, not here. **24 records rejected**, each with a reason in `state/rejected.jsonl` — the large majority environmental-science records surfaced by the broad MeSH sweep with no atmospheric particulate pathway (freshwater mercury, soil organic carbon, riverine methane, waterborne microplastics). **Six carried records have no abstract** in Crossref, Europe PMC or Semantic Scholar — four Elsevier, one Springer, one RSC — and are marked METADATA ONLY; a seventh, the RSC Delhi wavelet paper, has an abstract truncated mid-sentence at source. Fourth consecutive issue affected; the re-check remains a weekly sweep. **Three defects were closed in this issue.** First and most serious: `plots.py` carried a **silent hardcoded fallback for f6**, so an issue whose corpus JSON lacked a `LIFECOURSE` key rendered life-course counts `[5, 3, 2, 3, 4]` and exemplars (“firefighter biomarkers”, “care-home filtration”, “PAH endocrine axis”) drawn from issues weeks earlier. Today’s corpus contains none of those records, so the figure would have shipped numbers describing no paper in the issue. It was caught only by reading the rendered PNG against the store. `corpus.save()` now takes and writes `lifecourse`, and a missing key is a hard build failure rather than a plausible-looking placeholder. Second: the PubMed `efetch` parser was reading DOIs with an unscoped `.//ArticleId` XPath, which matched reference-list identifiers — **15 of 33 records** received a DOI belonging to a cited paper, several from the 2010s. The lookup is now scoped to `PubmedData/ArticleIdList` with an `ELocationID` fallback, and `check_dois.py` passes 0 fail / 0 warn. Third: the geography mapper fell through for *Faroe Islands*, *West Africa* and *Idealised terrain*, silently pooling all three into “Global / multi-region” and inflating that bar from 8 to 10; all three are now mapped. `design_group` did **not** fall through this issue — the first clean run in four — because every design label written on a record was chosen from the existing mapped vocabulary. The combined `f2_design_endpoint.png` panel is **not used in this issue**: with 14 distinct endpoint labels its y-axis text overlaps the donut centre and the architecture legend, so `f2a` and `f2b` are shown separately pending a layout fix. Subtopic, design, particle-metric and geography labels are assigned from title, abstract, keywords and MeSH; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Effect estimates on the forest plot are heterogeneous by metric and are not pooled: percentage excess risks are shown as $1 + \mathrm{ER}/100$ and percentage changes in continuous biomarkers as geometric-mean ratios, purely so they share a ratio axis. All DOI links resolve to the publisher of record. Content derived from PubMed, Crossref and Europe PMC metadata; abstracts remain the copyright of their respective publishers.